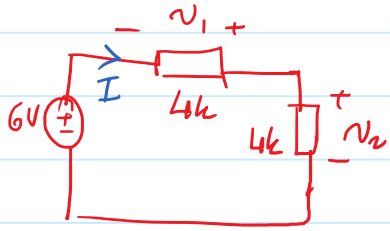
Q1 (10 pts)



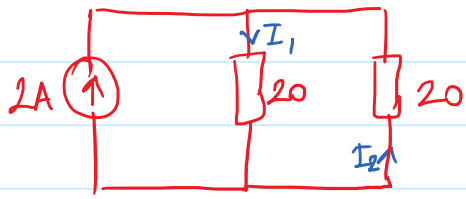
$$I = \frac{6}{8k}\ A$$

1pts 4pts

$$v_1 = -4k \cdot \frac{6}{8k} = \boxed{-3V = v_1} \quad (5pts)$$

$$v_2 = 4k \cdot \frac{6}{8k} = \boxed{3V = v_2} \quad (5pts)$$

Q2 (10pts)



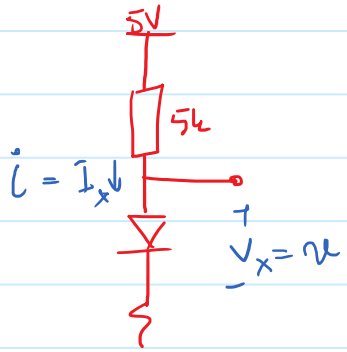
This is current divides

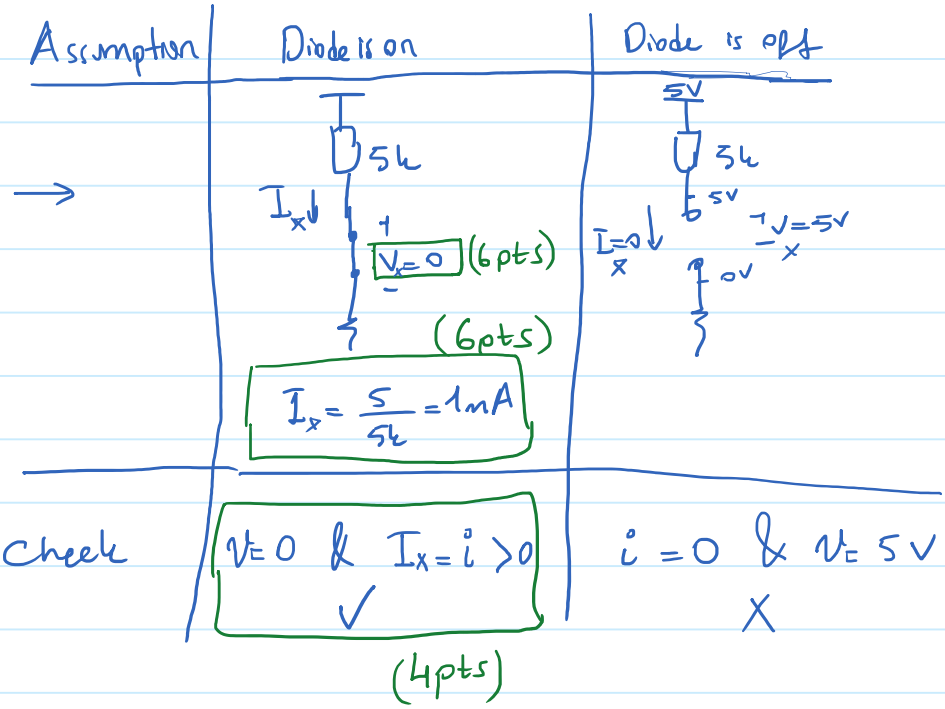
$I_1 = -I_2 = 1A$

1pts 4pts

$\boxed{I_1 = 1A}$ (5pts), $\boxed{I_2 = -1A}$ (5pts)

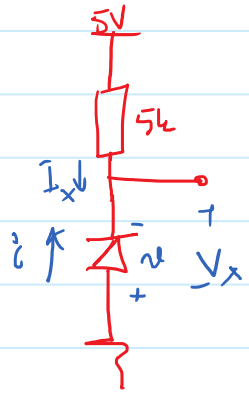
Q3 (16 pts)

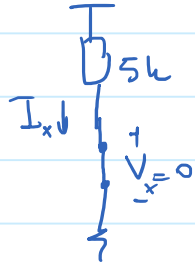
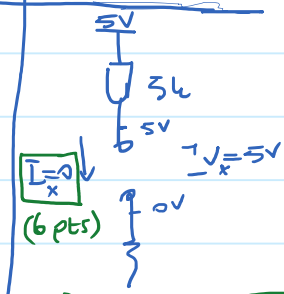

5V
5k
i = Ix
+
Vx = v
-


Assumption
Diode is on
Diode is off
5k
Ix
+
V=0
-
(6pts)
(6pts)
Ix = 5/5k = 1mA
5V
5k
0V
Ix = 0
+
V = 5V
-
x

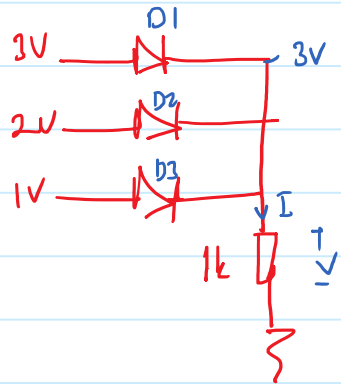
Check
V = 0 & Ix = i > 0
✓
i = 0 & V = 5V
X
(4pts)

Q4 (16 pts)



Assumption	Diode is on	Diode is off
→		
	$I_x = \frac{5}{5k} = 1mA$	$v = -V_x = -5V$ (6 pts)
Check	$v=0$ & $i=-I_x<0$	$i=0$ & $v=-5V$
	X	V (4pts)

Q5 (8 pts)



D1 is on, D2 and D3 are off.

Thus $V = 3V$ (4pts) & $I = \frac{3}{1k} = 3mA = I$ (4pts)

Check

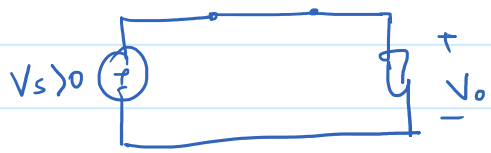
$V_{D1} = 0$, $I_{D1} = 3mA$ ✓

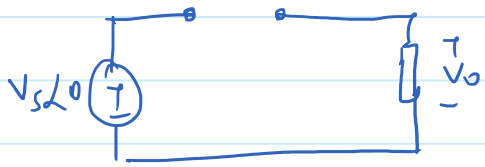
$V_{D2} = -1$, $I_{D2} = 0A$ ✓

$V_{D3} = -2$, $I_{D3} = 0A$ ✓

Q6 (15 pts)

If you have done something relevant, you get 5pts.



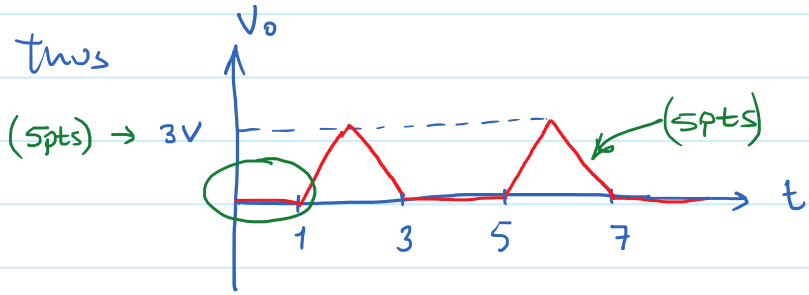


$V_s = V_o$

$V_o = 0$

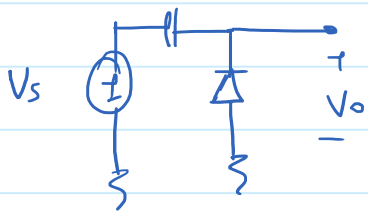
$$V_o = \begin{cases} V_s, & V_s \geq 0 \\ 0, & V_s < 0 \end{cases}$$



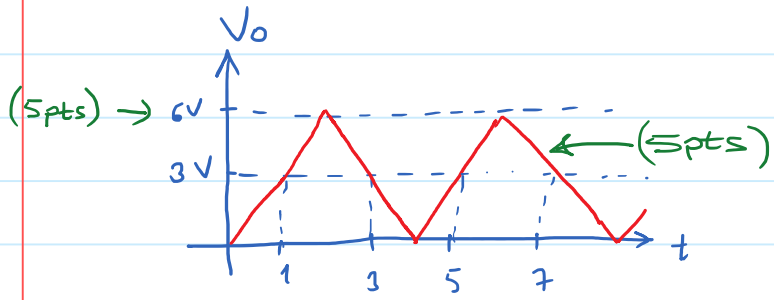


Q7 (15 pts)

If you have done something relevant, you get 5pts.

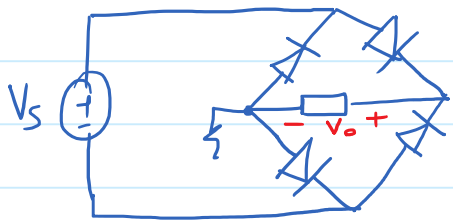


V_o cannot be at a negative potential. The diode will not let it happen. The V_s will be shifted upward such that minimum value of V_o will be at 0V.

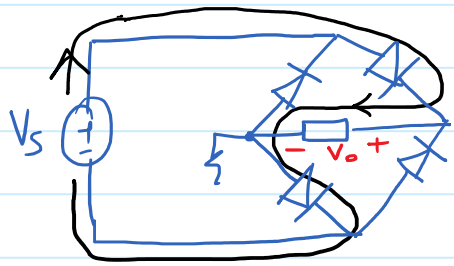


Q8 (10pts)

If you have done something relevant, you get 4pts.

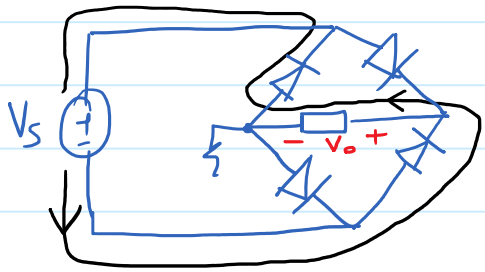


If $V_s > 0$



$V_o = V_s$

If $V_s < 0$



$V_o = -V_s$

$$V_o = \begin{cases} V_s, & V_s > 0 \\ -V_s, & V_s < 0 \end{cases}$$

Thus $V_o = |V_s|$

